Kyler Cao

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Education

Texas A&M University College Station, TX

Bachelor of Science in Computer Science, Minor in Business | ***GPA: 3.59/4.00*** *Aug. 2023 - May 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Computer Organization, Foundations of Software Engineering, Database Systems, Machine Learning

Experience

Associate Software Engineer Aug. 2026 - Present

Global Shop Solutions *Remote*

- Built an automation suite for migrating a legacy **COBOL ERP to C#** (∼**470K lines of COBOL** across 200 programs into ∼**2.3M lines of C#**) that runs **2,000+ UI regression test cases** in parallel across Windows test VMs, coordinated by an internal .NET/TypeScript tracker
- Building an internal **AI gateway** on **LiteLLM and AWS Bedrock** to replace an external AI provider hub, with per-caller usage tracking and a tested migration cutover for existing services
- Helped build a **support data extraction** feature for a **VB.NET/C#** desktop ERP, fixing a sign-in security flaw, adding CI, and moving its sign-in call off the UI thread to remove freezes of up to **240 seconds**

Product & Engineering Intern May 2026 - Aug. 2026

Global Shop Solutions *The Woodlands, TX*

- Developed components for **Launchpad**, an internal developer platform in **Go, PostgreSQL, and Kubernetes** that deploys applications declaratively through GitHub and Argo CD, with reconciled jobs, Keycloak authentication, and department-scoped access control
- Cut average ticket resolution time by **70%** with an **AI diagnosis dashboard** (React, Node.js, PostgreSQL) that correlates multimodal test evidence with structured model outputs to rank causes and draft reviewable diagnoses
- Built a **C#/.NET service** that compiles COBOL and C# projects on demand for a **20-person** engineering team

Internal Tools Developer Oct. 2025 - May 2026

Texas A&M TechHub *College Station, TX*

- Built a delivery management system (**Flask, React, MySQL**) that processes **200+ monthly orders** with auditable histories
- Shipped **8+ internal tools** for inventory automation, documentation, and asset lifecycle tracking across a **1,000+ device** campus fleet
- Deployed **5,000+ university devices**, cut average imaging time by **15%**, and resolved **200+ faculty and staff support cases**

Teaching Assistant Aug. 2024 - May 2025

Texas A&M University *College Station, TX*

- Taught **Python** to **120+ students** through labs, office hours, and weekly problem-solving reviews
- Graded **400+ assignments and exams** with written feedback in a **24-hour turnaround** and created exam-review materials

Projects

CodeHawk | *Go, Python, LangGraph, Redis Streams, pgvector* | Private Project June 2026 - Present

- Built the core of a tiered GitHub PR review system that has reviewed **50+ PRs**: a **Go API** for signed webhook ingestion, diff parsing, Redis Streams job dispatch, and PR publishing, plus a **Python/LangGraph worker** that runs the review agents
- Made reviews durable with a transactional outbox, immutable archives, idempotency, and retry/dead-letter handling, and documented a reproducible Docker Compose setup for local testing

Seer | *Rust, ratatui, crossterm, iroh, portable-pty* | Private Project Sept. 2026 - Present

- Built a multi-crate **Rust** terminal runtime for coding agents with a **ratatui/crossterm** TUI, persistent workspace and pane state, and local PTY runtimes
- Added brokered multi-user sessions with credentialed clients and end-to-end encrypted remote connections over **iroh**, covered by integration tests for handshakes, forwarding, detach, and lifecycle

Technical Skills

Languages: Rust, Python, Go, TypeScript/JavaScript, C#, SQL, Java, C++, COBOL
Frameworks & Libraries: React, .NET, Flask, Node.js, LangGraph, ratatui, crossterm, portable-pty, pandas, scikit-learn
Tools & Platforms: Git, Docker, Kubernetes, Argo CD, Linux, PostgreSQL, MySQL, Redis, iroh, Slurm